METRIC

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COMMERCIAL ITEM DESCRIPTION
THEODOLITES, SURVEYING, DIRECTION: SECOND ORDER

The General Services Administration has authorized the use of this Commercial Item Description (CID) for all federal agencies. The original issue of this CID replaced MIL-T-14132.

1. SCOPE. This CID covers two types of illuminated theodolites with accessories, cases, and tripods.

2. CLASSIFICATION. The theodolite will be one of the following types, as specified (see 7.3):

Type I - Auto-reading digital direction-type
Type II - Optical-reading direction-type
Class 1 1 second
Class II 0.002 mil

3. SALIENT CHARACTERISTICS.

3.1 Description. Theodolites shall be a direction-type, capable of second order accuracy (see 7.7), and capable of displaying readings in seconds and mils. Each theodolite, including the telescope, horizontal and vertical circles, circle reading system, leveling device, integral illumination system, and tribrach, shall weigh not more than 8.0 kg. Each theodolite shall be furnished with a carrying case, a theodolite cover, and when specified, a transport case, a battery and lighting accessory case, tools and accessories, and a tripod.

Beneficial comments, recommendations, additions, deletions, clarifications etc., and any other data which may improve this document should be sent to: ENGINEER RESEARCH AND DEVELOPMENT CENTER, CORPS OF ENGINEERS, U.S. ARMY TOPOGRAPHIC ENGINEERING CENTER, ATTN CEERD-TS, 7701 TELEGRAPH ROAD, ALEXANDRIA, VA 22315-3864.

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3.2 Theodolite characteristics. Theodolites shall be an auto-reading digital or an optical-reading direction-type.

3.2.1 Telescope. The telescope shall be mounted so that pointings may be made to the zenith by use of an elbow eyepiece. The telescope shall have a means for illuminating the reticle and a means for controlling the illumination. The stadia ratio shall be 1/100. The telescope shall display an erect image. The reticle shall have a pattern similar to figure 1.

3.2.1.1 Telescope eyepiece. The telescope eyepiece shall contain a diopter scale plainly marked in plus or minus units and shall have an index mark referencing the setting.

3.2.1.1.1 Elbow eyepiece. Unless otherwise specified (see 7.3), the elbow eyepiece shall enable an operator to observe vertical angles to the zenith and to observe an azimuth mark, with the telescope in either the direct or reverse position. The eyepiece shall give an erect image of the reading scales.

3.2.1.1.2 Solar filter. Unless otherwise specified (see 7.3), a solar filter shall be provided that shall fit both the telescope eyepiece and the elbow eyepiece.

3.2.1.1.3 Auto-collimation eyepiece. When specified (see 7.3), an auto-collimation eyepiece shall be furnished. The eyepiece shall be interchangeable with the standard eyepiece. The magnification of the telescope with the auto-collimation eyepiece installed shall be not less than 18 diameters.

3.2.2 Optical plummet. The optical plummet shall produce a clear image throughout the operating range, from tripod height to not less than 30.5 meters.

3.2.3 Circle identification. The images of or readouts from the horizontal and vertical circles shall be identified.

3.2.4 Microscope. For type II, a microscope shall be provided to view the horizontal and vertical circles and breakdown scales, simultaneously. The images of the horizontal and vertical circles, index lines and breakdown scales shall appear erect and in the same plane. The graduation interval of the micrometer shall be 1 second for class I circles, or 0.002 mil for class II circles.

3.2.5 Microprocessor and control/display panel (for type I only).

3.2.5.1 Microprocessor. The microprocessor shall have a nonvolatile memory with sufficient capacity to store all data, constants, operating parameters, and sufficient random access memory to perform required data reductions.

3.2.5.2 Control/display panel. The panel shall contain all function keys, an alpha-numeric display unit, and a means to record and process data. All keys shall be clearly identified as to function. Receptacles shall be labeled according to function. Warning and indicator lights required for operation of the instrument shall also be located on the panel. The panel shall automatically display observed horizontal and vertical angles at a precision and repeatability required to satisfy second order accuracies.

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3.2.6 Levels. The instrument shall be equipped with a leveling capability which allows horizontal and vertical angle measuring at the precision and repeatability required to satisfy second order accuracies.

3.2.6.1 Vertical circle indexing. For type II, vertical circle indexing shall be controlled by a self-indexing device, unless an altitude level is specified (see 7.3). When the vertical circle in the direct position reads 90 degrees or 1600 mils, the telescope shall be horizontal.

3.2.6.2 Self-indexing device. The self-indexing device for type II theodolites shall be damped so that oscillation is stopped in not greater than 5 seconds of time. The device shall be accurate to within 10 seconds of arc or 0.05 mil.

3.2.6.3 Altitude level. On type II theodolites, when an altitude level is specified (see 7.3), the vertical circle shall be ungraduated but shall have a sensitivity 30 seconds, ±5 seconds for class I, and 0.15 mil, ± 0.03 mil for class II, for each 2 mm.

3.2.7 Tribrach. The tribrach shall consist of the leveling screws, base plate, and a device for locking the theodolite assembly. The base plate shall have at its center a 5/8-inch nut of the unified coarse thread series (0.625-11 UNC-2B) for compatibility with existing equipment.

3.2.8 Illumination. Illumination shall be by natural and artificial means.

3.2.8.1 Natural illumination. The theodolite shall be readable in all daylight conditions. For type II theodolites, natural illumination shall be useable at any orientation.

3.2.8.2 Artificial illumination. No light shall be visible beyond a 30.5 meter circle with the instrument as center point, under moonless and starless night conditions. All externally visible light produced shall be intensity adjustable.

3.2.8.3 Handlamp. For type II theodolites, when specified (see 7.3), a handlamp shall be provided and shall have connecting cord not less than 1.5 meters long.

3.2.8.4 Battery and lighting accessory case. For type II theodolites, when specified (see 7.3), a case that shall not be damaged when dropped from carrying height, at any operating temperature specified herein, shall be provided. The case shall provide mounting for batteries that shall provide power for the handlamp and theodolite. The case shall provide space for storage of the hand lamp, connector cords, and any spares. The case shall have a provision for mounting on the tripod.

3.2.9 Power System

3.2.9.1 Power. Theodolites shall operate from an internal or external battery and shall contain built-in protection against reverse polarity from the power source, including a battery compartment that precludes improper battery insertion.

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3.2.9.2 System power. System power, through multiple battery packs, shall be provided to support not less than 500 angle measurements and provide no less than eight hours of measuring time at 20° C.

3.2.9.3 Auxiliary power. When specified (see 7.3), type I theodolites shall also operate from external 12-volt DC batteries, providing the same operating capability as with the internal power source. An external power cable shall be provided with the system that shall not be less than 6 meters in length and shall terminate in automotive type battery clamps which are identified by a + and -.

3.2.9.4 Battery charger. System batteries shall be rechargeable using a 110/220 volt, 50/60 Hz charger that shall be supplied with the system. The battery charger shall be capable of concurrently charging at least two battery packs or a built-in battery charger shall be furnished with each battery pack. The battery shall not be damaged nor lose capacity when charged continuously from the charger.

3.2.10 Cases and covers.

3.2.10.1 Theodolite cover. A lightweight, flexible, waterproof cover shall be furnished to protect the theodolite from rain and dust. The cover shall withstand the environmental conditions specified herein without damage.

3.2.10.2 Carrying case. A carrying case to protect the theodolite from damage shall be provided. The case shall contain a desiccant adequate to the size of the container and provide for storage of ancillary items. The theodolite shall be fastened within the case to prevent it falling out when the case is opened. The weight of the carrying case with ancillary items shall not exceed 5.9 kg.

3.2.10.3 Transport case. When specified (see 7.3), a reusable transport case that will house and protect the theodolite in its carrying case, during transport, shipment, and storage shall be provided. The case shall be of rigid, sturdy construction, and shall weigh not more than 6.8 kg. The case shall have removable inserts contoured to fit snugly around the carrying case. The case shall have carrying handle(s) and lock(s) or clamp(s). All hardware shall be corrosion resistant or treated to resist corrosion.

3.2.10.4 Backpack. When specified (see 7.3), a backpack shall be provided for transporting the theodolite in its carrying case. Shoulder straps shall be cushioned or padded. The backpack shall be fitted with pockets for carrying accessories. A cover shall fit snugly over the top of the backpack to prevent penetration of rain. The color of the backpack shall be army green.

3.2.11 Winterization kit. Unless otherwise specified (see 7.3), a winterization kit shall be furnished for operation in temperatures below -20°C. The winterization kit at a minimum shall include covers for all control knobs, as required, and leveling screws, as well as shields for any telescope, microscope, and optical plummet eyepieces. A winterization kit for type II theodolites shall be capable of warming the instrument to operating temperature from -32°C within 25 minutes and from -43°C within 45 minutes.

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3.2.12 Treatment and painting. Unless otherwise specified, (see 7.3), the color shall be lustreless army green. Untreated, exposed, exterior surfaces of the theodolite, cases, and accessories shall be treated and painted the same as those items the manufacturer offers for sale commercially.

3.2.13 Tripod. When specified (see 7.3), a collapsible type telescoping leg tripod conforming to A-A-52298 shall be furnished with each theodolite.

3.2.14 Tools and accessories. When specified (see 7.3), the following items shall be furnished with each theodolite:

Item	Quantity
Lens brush	1
Chamois skin (20 x 20 cm)	1
Container of lubricant	1
Recommended tools	1 set
Lens tissue	1 pkg
Orangewood sticks (commercial grade)	1 pkg
Manual, theodolite instruction, with protective cover, (contractor's standard)	1

The contractor may add items to this list as required for field maintenance and adjustments. All tools and accessories shall be contained in a case.

3.3 Environmental requirements.

3.3.1 Operating temperature range. The instruments shall determine azimuths required for second order accuracy within an ambient temperature range of -40°C to 52°C.

3.3.2 Storage temperature range. The instrument shall not be damaged by storage at temperatures as low as -46°C for 6 hours and as high as 68°C for 4 hours, after stabilization (see 7.5.3), and without solar radiation.

3.3.3 Solar radiation. The instruments shall operate as specified herein and shall not be damaged by the full impact of solar radiation of 355 British Thermal Units (BTU) per square foot per hour (BTU/ft^2/hr) at an air temperature of 52°C for 4 hours.

3.3.4 Humidity. The instruments shall operate as specified herein and shall not be damaged when subjected to relative humidity of at least 95 percent at air temperatures from -32°C to 43°C. The instrument shall be exposed for not less than 24 hours after temperature stabilization.

3.3.5 Fungus and moisture resistance. Electrical circuitry and components that are susceptible to fungus and moisture damage shall be hermetically sealed or otherwise coated for protection.

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3.3.5 Rainfall.

3.3.5.1 Assembled instrument. When assembled and in the normal operating position, the instrument shall not be damaged or show any signs of moisture penetration after exposure to wind driven rainfall of 8 ±2 cm per hour, from any direction, and at any angle from vertical to 45 degrees from vertical, for a period of not less than 5 minutes.

3.3.5.2 Cases. The carrying case and transport case shall each meet the rainfall requirements of 3.3.5.1 for a period of not less than one hour.

3.3.6 Blowing dust. The instruments shall endure a blowing dust environment in the storage mode with the following criteria:

a. The air velocities shall be 1.5 to 8.9 m/s.
b. The dust composition shall be consistent with red china clay or silicon flour.
c. The dust concentration shall be maintained at 10.6 ±7 g/m^3.
d. The instrument shall be oriented so that all surfaces shall face the blowing dust for 90 minutes per surface.
e. The air temperature shall be 23°C for 6 hours , and 50°C for 6 hours.

3.3.7 Blowing sand. The instruments shall endure a blowing sand environment in the storage mode with the following criteria:

a. The air velocities used shall be 18 to 29 m/s.
b. The suggested sand composition (at least 95% silicate oxide by weight) is as follows:
   1.00% shall be retained by a 20 mesh screen.
   2.00% shall be retained by a 30 mesh screen.
   15.00% shall be retained by a 40 mesh screen.
   37.00% shall be retained by a 50 mesh screen.
   30.00% shall be retained by a 70 mesh screen.
   15.00% shall be retained by a 100 mesh screen.
c. The sand concentration for natural conditions shall be 2.2 ±0.50 g/m^3.
d. The instruments shall be oriented so that all surfaces shall face the blowing sand for 90 minutes per surface.
e. The air temperature shall be conducted at 50°C.

3.3.8 Vibration. The instrument shall be constructed so that parts do not become loose, permanent settings do not shift, and no degradation in performance is caused when stored in transport cases and subjected to 45 minutes of vibration in each of the three axes. The test bed shall produce a 2.54 cm, double amplitude, orbital path at 5 Hz .

3.3.9 Shock. The instrument shall be constructed so that parts do not become loose, permanent settings do not shift, and no degradation in performance is caused when stored in transport cases and subjected to drops of 33 cm as follows: each of the 6 faces, each of the 8 corners, and each of the 12 edges, for a total of 26 drops.

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3.3.10 Electromagnetic interference. Type I theodolites shall have design characteristics for the control of electromagnetic interference emission and susceptibility.

3.3.10.1 Conducted emissions. Conducted emissions (CE) on power leads and returns, that obtain power from sources not part of the unit shall not exceed the values in figure 2 for a frequency scan from 10 kHz to 10 MHz.

3.3.10.2 Conducted susceptibility. Conducted susceptibility (CS) on all interconnecting cables, including power cables, shall not exceed the values in figure 3 for a frequency spectrum scan from 10 kHz to 10MHz. Use curve A for frequency ranges 10 kHz to 2MHz, and curve B for frequency ranges from 2MHz to 10MHz.

3.3.10.3 Radiated emissions. Radiated emissions (RE) from the instrument and all interconnecting cables shall not exceed the values in figure 4 for a frequency spectrum scan from 2 MHz to 18 GHz. Testing is required up to 1 GHz or 10 times the highest intentionally generated frequency within the instrument, whichever is greater. Measurements beyond 18 GHz are not required.

3.3.10.4 Radiated susceptibility. Radiated susceptibility (RS) shall not exceed 50.00 volts/meter for a frequency spectrum scan from 2 MHz to 18 GHz. The requirements shall be met for vertically polarized fields up to 30 MHz and for both horizontally and vertically polarized fields above 30 MHz. Circular polarized fields are not acceptable.

4. REGULATORY REQUIREMENTS.

4.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

5. QUALITY ASSURANCE PROVISIONS. The supplier shall comply with the following for certification and acceptability.

5.1 Product conformance. The products provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market. The Government reserves the right to require proof of such conformance.

5.2 Market acceptability. The instrument offered shall be of the latest model of the standard product of the supplier and that has been used in the field for at least one year by commercial organizations.

5.3 Examination. Each instrument shall be examined for compliance with the requirements specified in 3.2 through 3.4. Any modification necessary following failure to meet the specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all examinations of performance and dimensional requirements. Non-compliance with any specified requirement, or the presence of one or more defects lessening maximum efficiency shall constitute cause for rejection.

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6. PACKAGING. Preservation, packing, and marking shall be as specified in the contract or order.

7. NOTES

7.1 Intended Use. Theodolites displaying in minutes and seconds are primarily intended for second-order survey and azimuth work. Theodolites displaying in mils are primarily intended for artillery and missile surveys.

7.2 When this CID is used for procurement the product conformance clause should appear in the solicitation.

7.3 Acquisition Requirements.

a. Title, number, and date of this document.
b. Type and class of theodolite required (see 2).
c. When a telescope elbow eyepiece is not required (see 3.2.1.1.1).
d. When a telescope solar filter is not required (see 3.2.1.1.2).
e. When a telescope auto-collimation eyepiece is not required (see 3.2.1.1.3).
f. When an altitude level is required (see 3.2.6.3)
g. When a handlamp is required (see 3.2.8.3)
h. When a battery lighting and accessory case is required (see 3.2.8.4)
i. When auxiliary power is required (see 3.2.9.3)
j. When a transport case is required (see 3.2.10.3)
k. When a backpack is required (see 3.2.10.4)
l. When a winterization kit is not required (see 3.2.11)
m. Color required (see 3.2.12)
n. When a tripod is required (see 3.2.13)
o. When tools and accessories are required (see 3.2.14)
p. Level of preservation and packing required (see 6).

7.4 Source of Documents. A-A-52298 is available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Ave, Philadelphia, PA 19120-5099.

7.5 Definitions. The following definitions apply for this document.

7.5.1 Mil. Mil is a military unit of measure equal to the angle subtended by 1/6400 of a circumference.

7.5.2 Recovered materials. For the purpose of this requirement, recovered materials are those that have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. Used, rebuilt, or remanufactured components, pieces and parts shall not be incorporated in the instrument.

7.5.3 Stabilization. Stabilization is a 2°C differential between the ambient temperature and the instrument.

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7.6 Subject term (key word) listing.

transits
electronic total station
control survey
missile survey
optical reading

7.7 Second-order accuracy. Information regarding second-order accuracy can be found in Standards and Specifications for Geodetic Control Networks, DMS NO. ST 031, available from the National Geodetic Information Branch (N/CG17x2), NOAA, Rockville, MD 20852.

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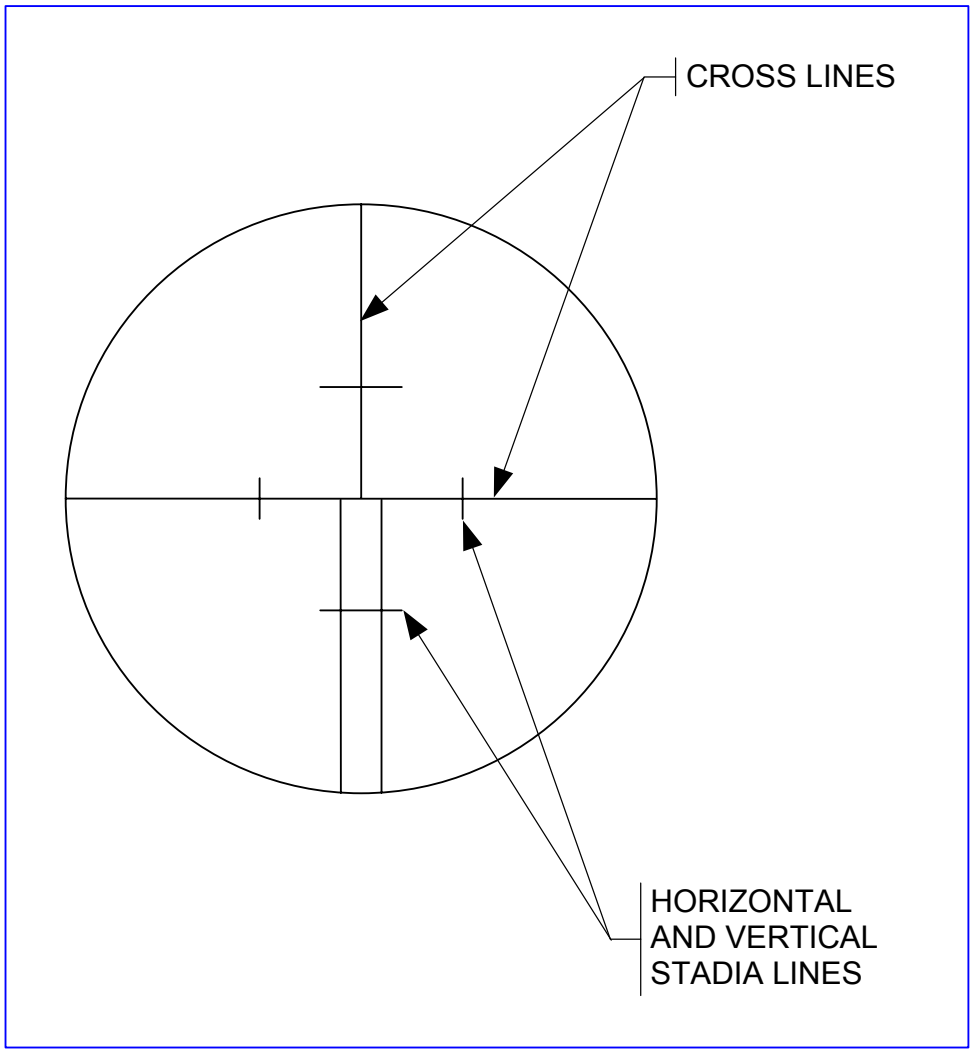


Figure 1. Reticle pattern for theodolite.
(Illustrative example only)

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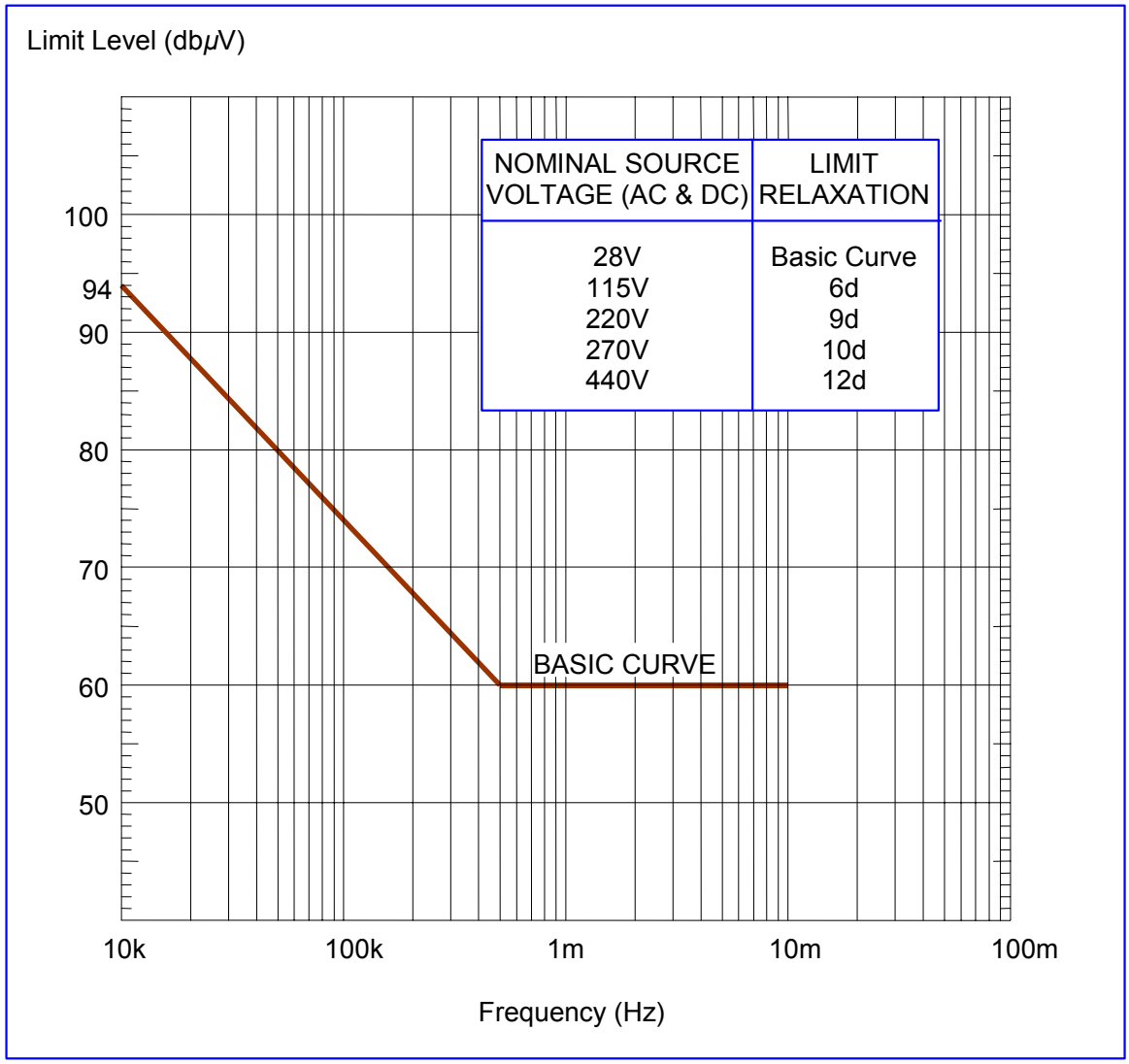


Figure 2. Conducted emissions limit

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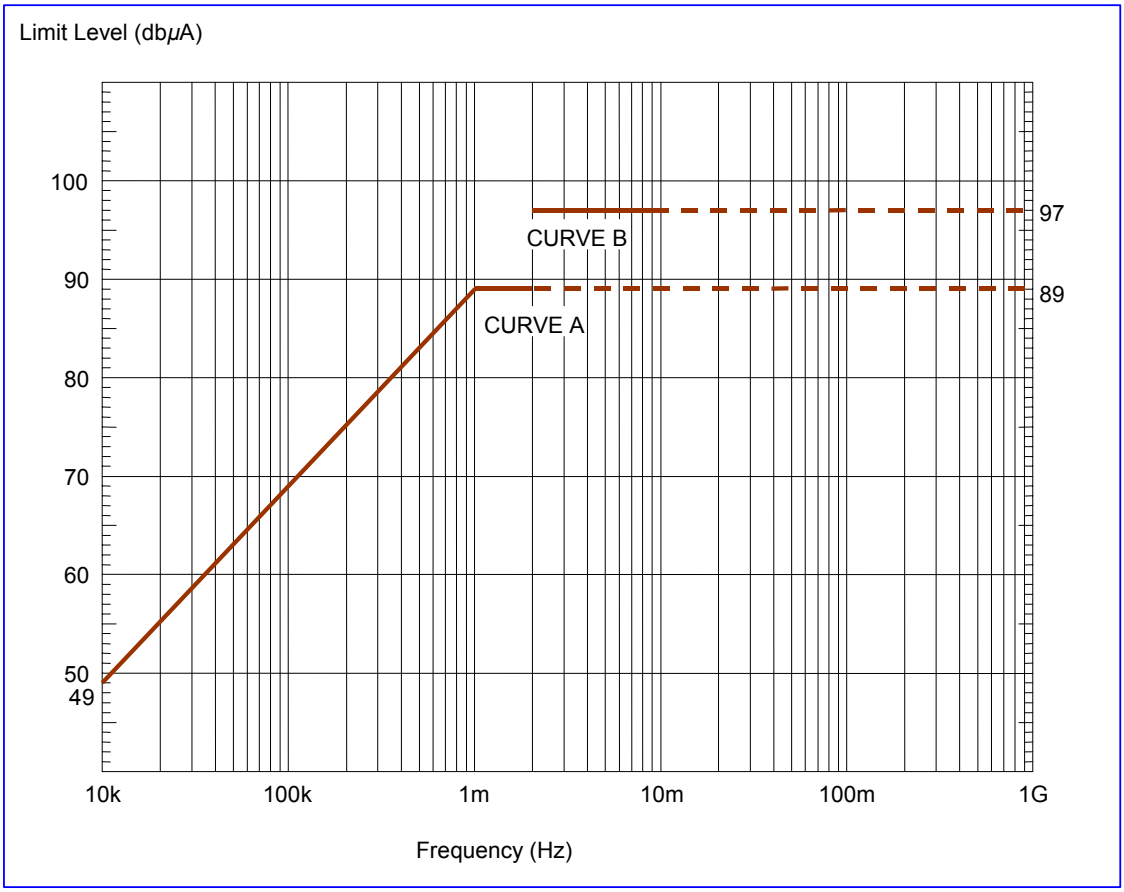


Figure 3. Conducted susceptibility limits.

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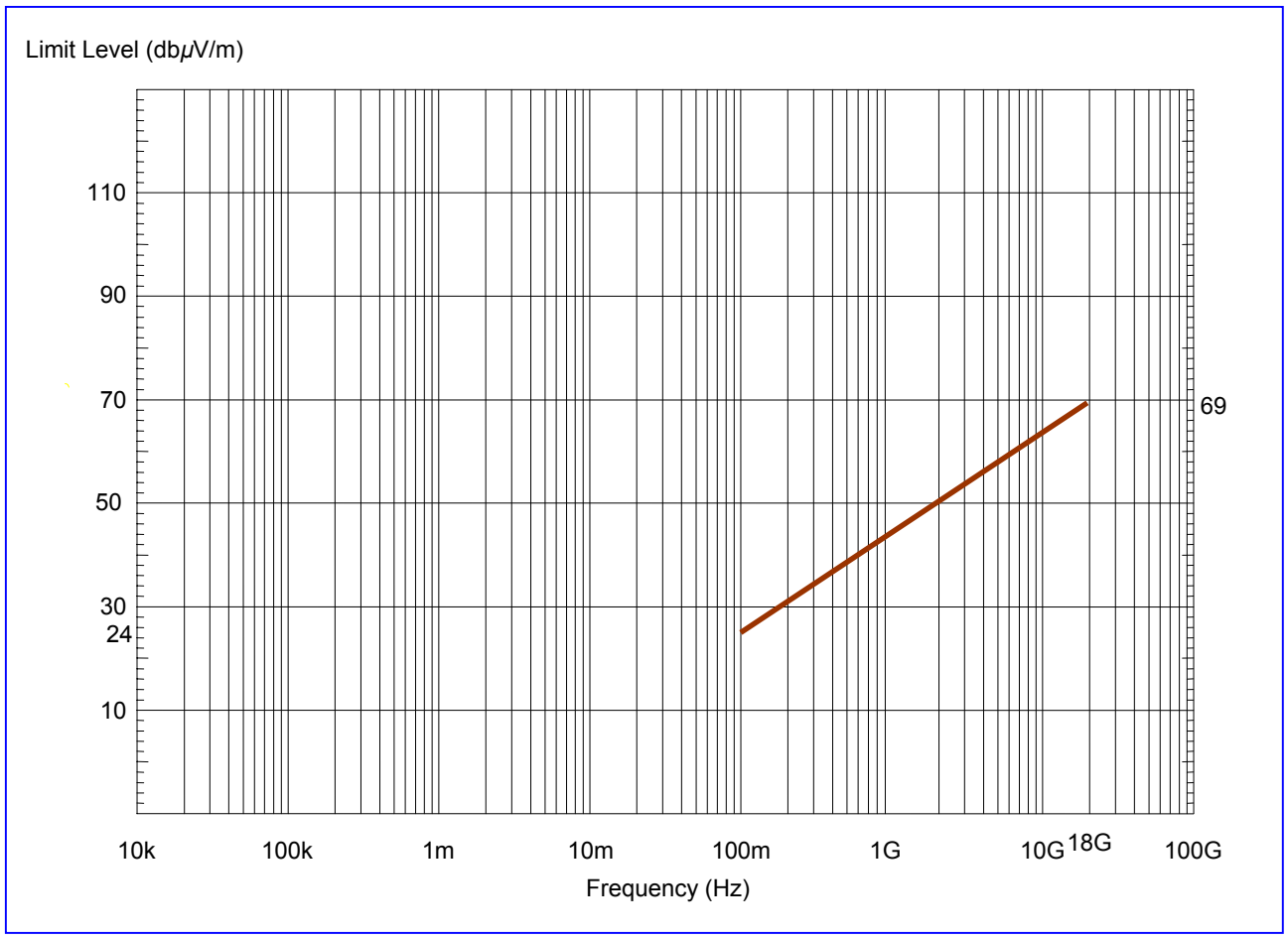


Figure 4. Radiated emission limit

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